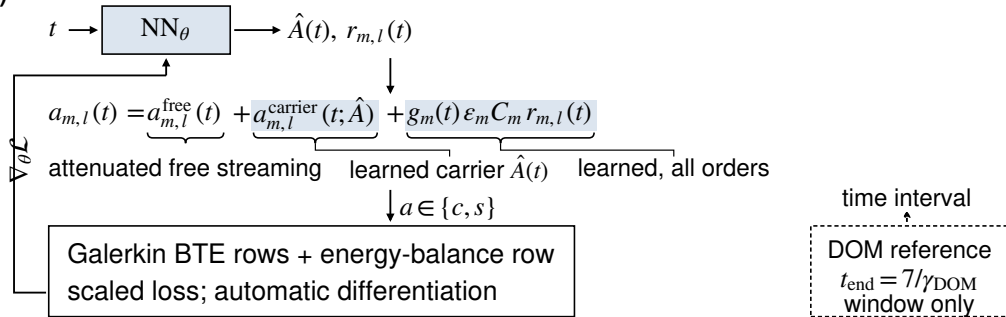


(a)

learned — functions of  $t$  alone; exact cosine IC at  $t=0$

$$f_m(x, \mu, t) = \sum_l \left[ \underbrace{c_{m,l}(t) \cos qx + s_{m,l}(t) \sin qx}_{\substack{\text{closed form in } x \\ \text{one harmonic, } q=2\pi/L \\ \text{periodic BC exact}}} \right] \underbrace{P_l(\mu)}_{\substack{\text{closed form in } \mu \\ \text{Legendre, } l \leq 8 \\ \text{moments algebraic}}}$$

(b)



Galerkin rows through  $l=8$ ; free-streaming closure at  $l=9$

time collocation on the reference-set interval; no solution labels or waypoints